

**Sri Lanka Institute of Information
Technology
Faculty of Computing**



EduFrame

Web-Based Video Browsing System

Web-Based Platform for Managing and Delivering Educational Video Content

Group – Y2-S1-MLB-B4G1-06

Year 2, Semester 1 – 2026

Lab Sheet 05 – Sequence Diagram

IT Number	Name
IT25101841	De Silva L. C. A.
IT25102695	Munasinghe M. H. V. S.
IT25103702	Bandara W. G. M. D.
IT25101767	Withana T. T.
IT25103673	Wickramathunga P. P.
IT25100859	Adikari A. M. T. D.

Individual Use Case Scenarios

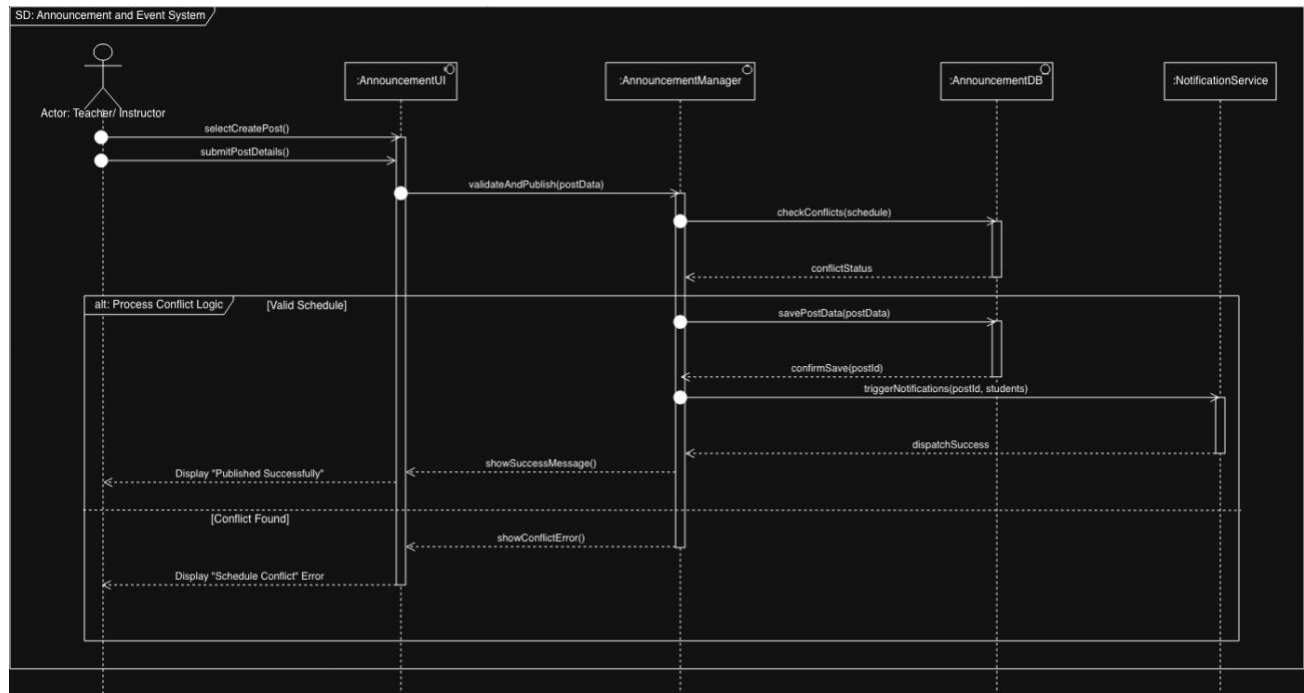
1. Announcement & Event System

Assigned to: De Silva L. C. A. (IT25101841)

Use Case Name	Publish & Manage Announcement / Event
Primary Actor(s)	Teacher / Instructor, Admin
Secondary Actor(s)	Student (receives the notification)
Precondition	The teacher or admin is logged in with permission to publish announcements.
Trigger	A teacher schedules a live webinar, or the admin needs to broadcast a platform-wide notice.
Postcondition	The announcement/event is visible to the relevant users, and notifications are dispatched.

Main Flow of Events:

1. Teacher/Admin selects 'Create Announcement' or 'Create Event'.
2. User enters the title, description, and (for events) the date, time, and venue/link.
3. System validates the schedule and does not conflict with existing published events for that course.
4. System publishes the announcement/event and triggers notifications to the relevant students.
5. Students view the announcement/event on their dashboard and receive a reminder notification as the date approaches.
6. Teacher/Admin can update the schedule, description, or venue, or remove expired/cancelled announcements and events.



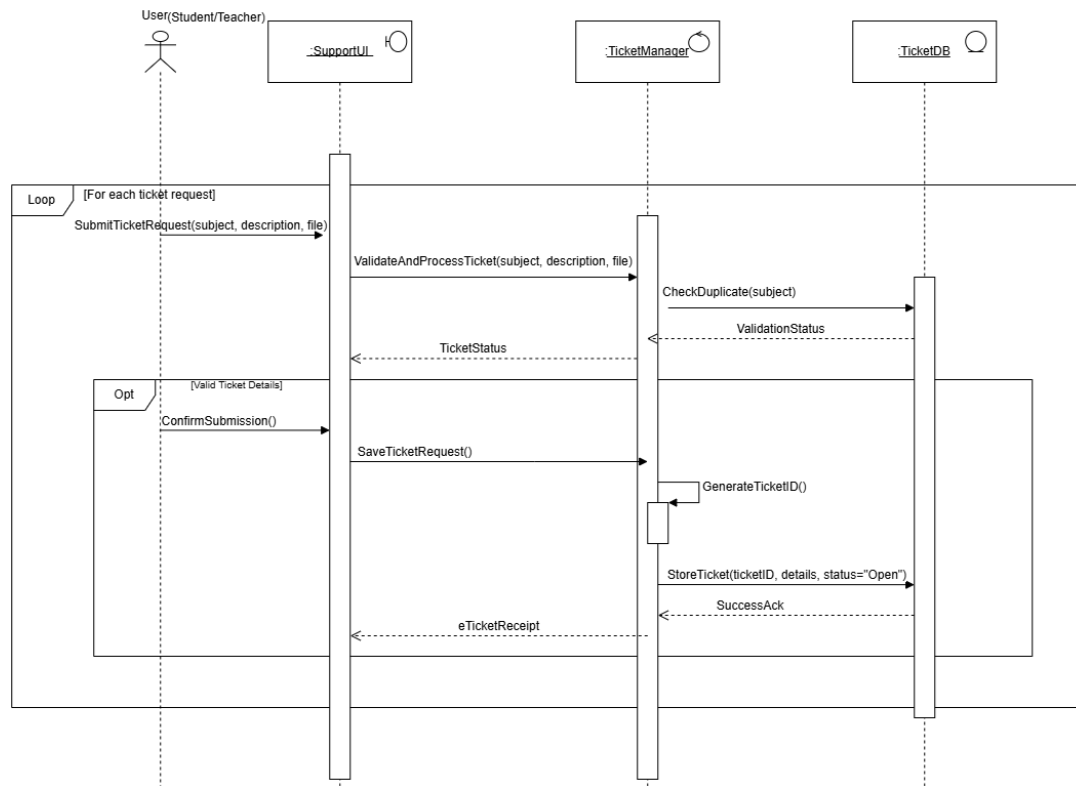
2. Ticketing System

Assigned to: Munasinghe M. H. V. S. (IT25102695)

Use Case Name	Submit & Manage Support Ticket
Primary Actor(s)	Student / Teacher (submit), Admin (manage)
Secondary Actor(s)	None
Precondition	The user is logged in and has encountered a technical issue or general inquiry.
Trigger	The ticket is recorded in the system with a status, and the user can track it until it is resolved.
Postcondition	The ticket is recorded in the system with a status, and the user can track it until it is resolved.

Main Flow of Events:

1. User navigates to 'Support' and selects 'Submit Ticket'.
2. User enters a subject, description of the issue, and optionally attaches a screenshot.
3. System creates the ticket with the status of 'Open' and a unique reference number.
4. Admin views the incoming ticket queue and opens the ticket to investigate.
5. Admin updates the ticket status to 'In Progress' and posts a response or requests more information.
6. Once resolved, admin updates the status to 'Resolved' and the user is notified.
7. User can view the status and full history of all tickets they have submitted at any time.



3. Advertisement Promotions

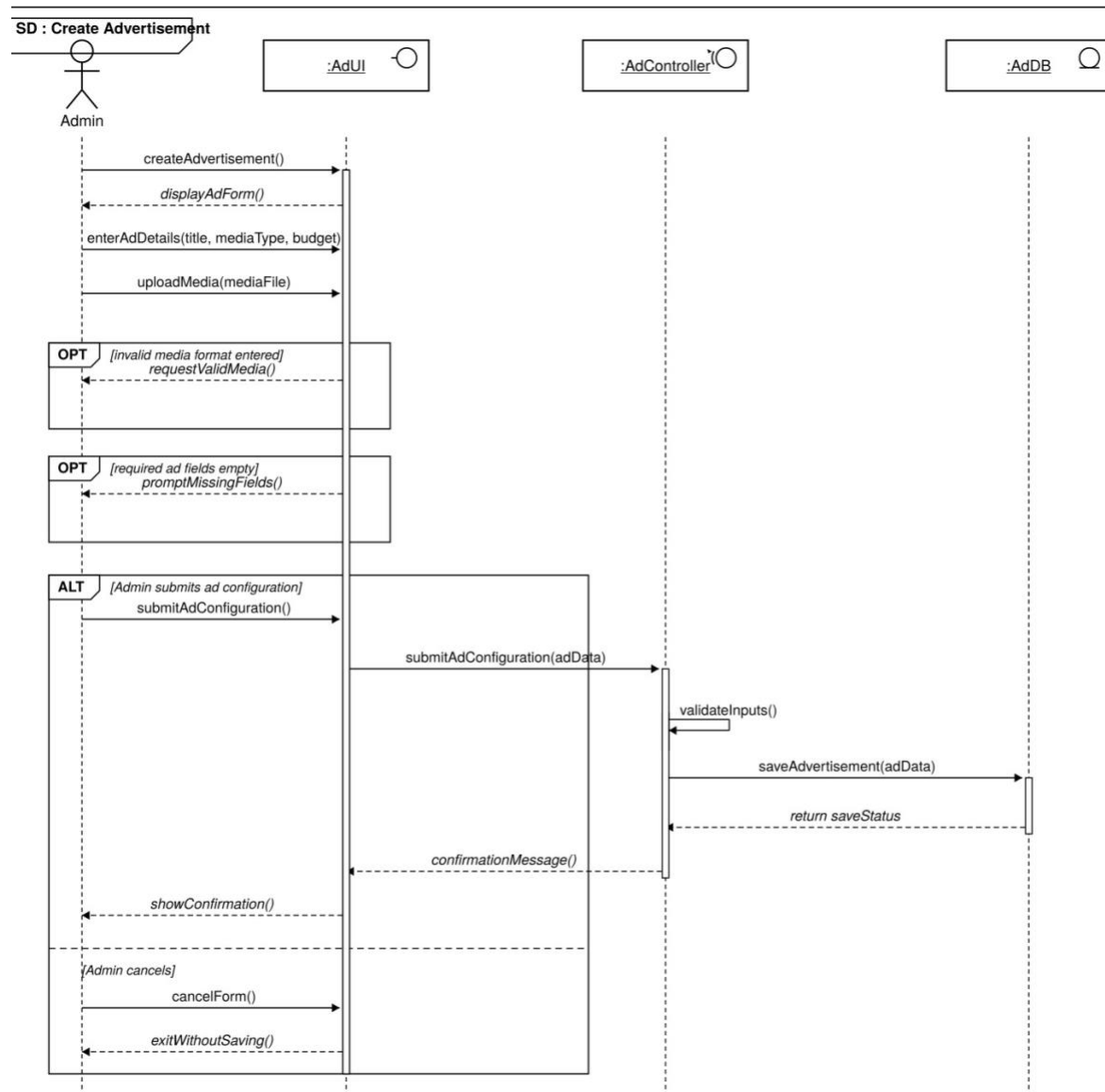
Assigned to: Bandara W. G. M. D. (IT25103702)

Use Case Name	Manage Advertisement
Primary Actor(s)	Admin
Secondary Actor(s)	All Users (view the resulting advertisement)
Precondition	The admin is logged in with administrative privileges.
Trigger	Admin wants to promote a featured course or platform announcement.
Postcondition	The advertisement is created, updated, or removed, and active advertisements are visible to users within their configured display period.

Main Flow of Events:

1. Admin logs in and navigate to the Advertisement Management screen.
2. Admin selects 'Create Advertisement' and enters the title, description, image, and the display start/end dates.
3. System validates the dates and saves the advertisement with status 'Scheduled' or 'Active'.
4. System automatically displays the advertisement to users on the homepage while the current date falls within its display period.

5. Admin can edit the advertisement's details, image, dates, or status at any time.
6. Admin can delete an advertisement, immediately removing it from display.



4. Course & Curriculum Management

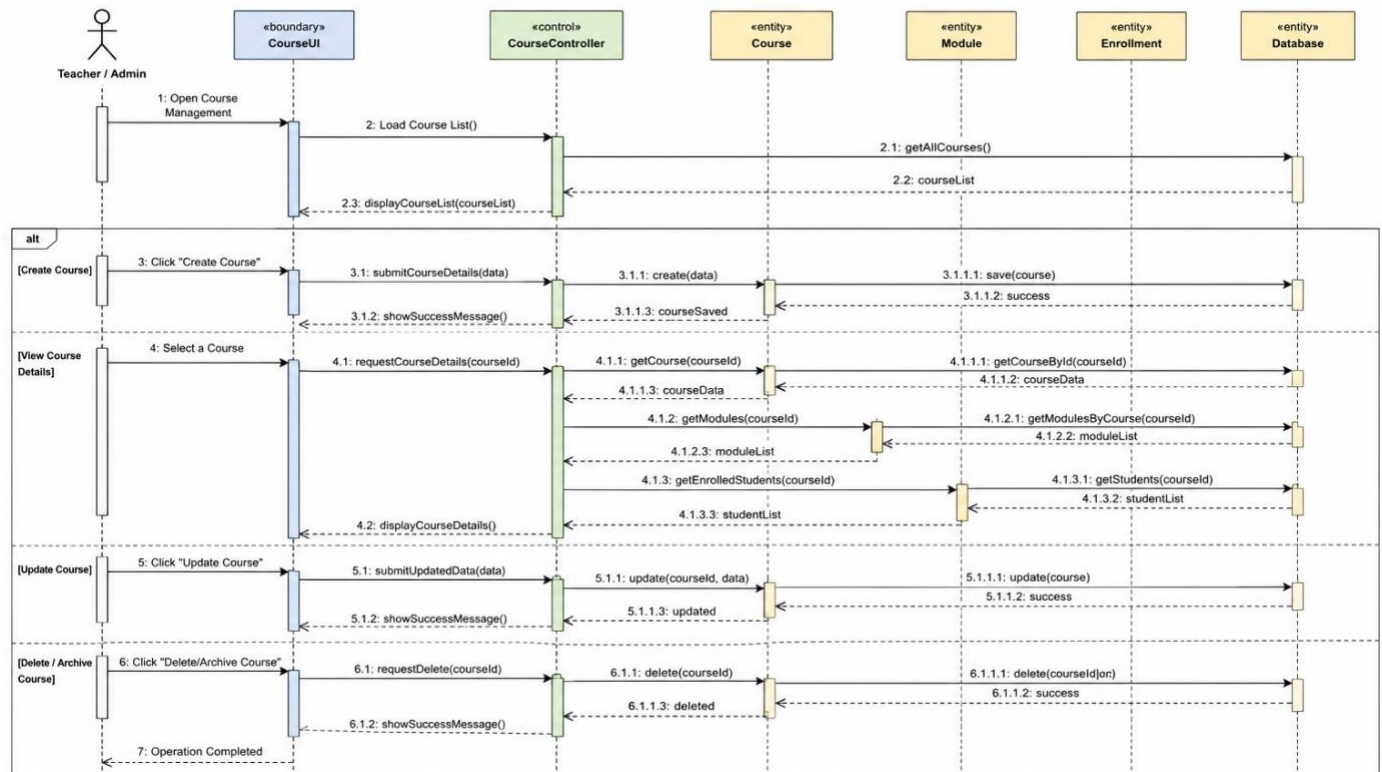
Assigned to: Withana T. T. (IT25101767)

Use Case Name	Enroll in Course / Manage Course
Primary Actor(s)	Student (enroll), Teacher / Instructor (manage)
Secondary Actor(s)	Admin (oversees the course catalogue)
Precondition	Student browses the course catalogue and chooses to enroll, or a teacher opens the course management screen.

Trigger	Student browses the course catalogue and chooses to enroll, or a teacher opens the course management screen.
Postcondition	For enrolment: the student is added to the course roster, and the course appears on their dashboard. For management: the course catalogue reflects the teacher's changes.

Main Flow of Events:

1. Teacher logs in and selects 'Create Course', entering title, description, category, and syllabus modules.
2. System validates the input and publishes the course to the catalogue.
3. Student searches or browses the catalogue and opens the course details page.
4. Student clicks 'Enroll'; system checks if the student is not already enrolled.
5. System adds the student to the course roster and updates the student's dashboard with the new course.
6. Teacher can later update the course information, syllabus, or cover image, or archive the course when it is no longer offered.



5. Video Management

Assigned to: Wickramathunga P. P. (IT25103673)

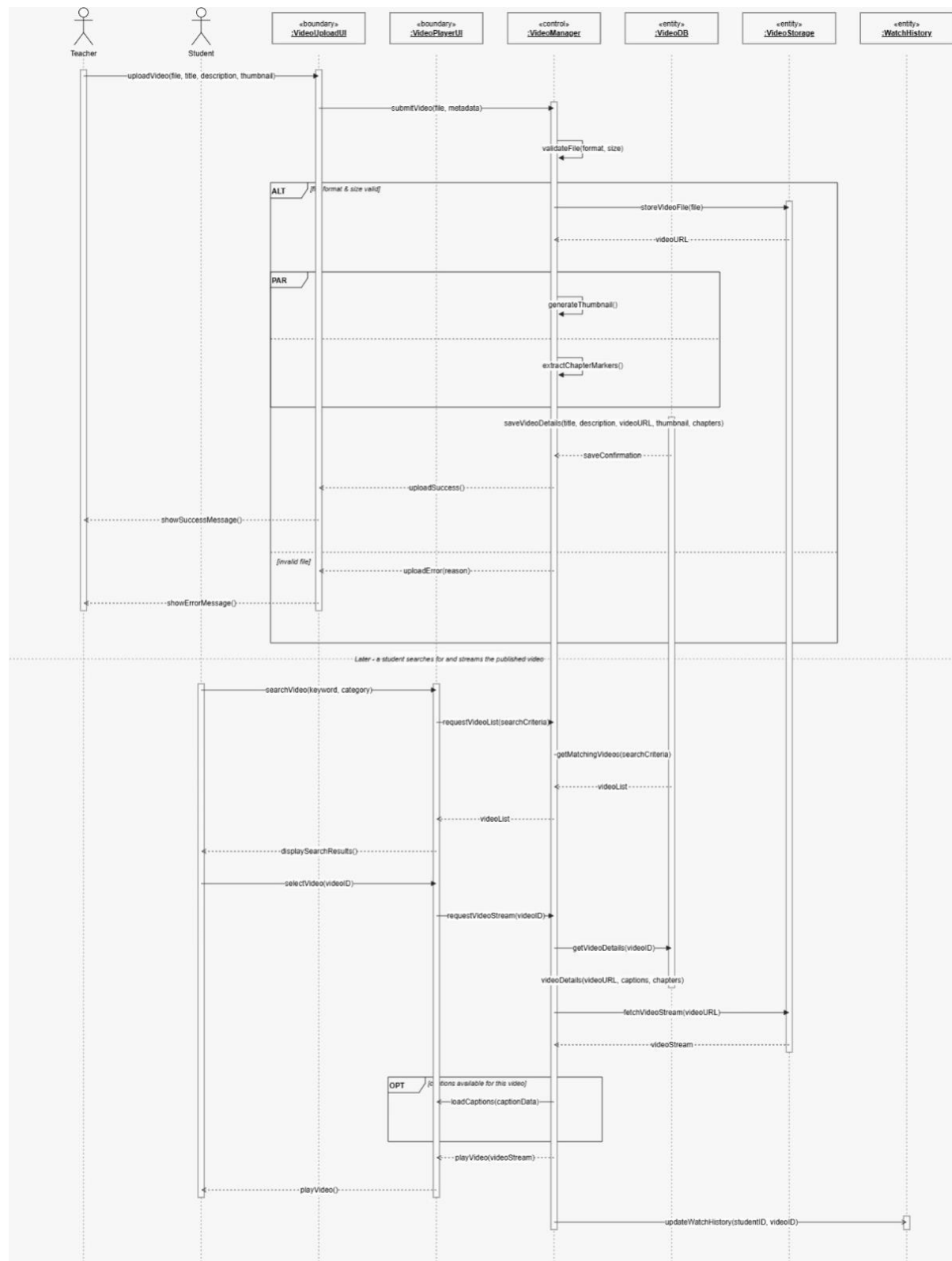
Use Case Name	Upload & Manage Video
Primary Actor(s)	Teacher / Instructor
Secondary Actor(s)	Student (watches the video), Admin (moderates' content)
Precondition	The teacher is logged in and owns or is assigned to the target course.
Trigger	The teacher wants to add a new lesson video to a course.
Postcondition	The video is stored, associated with the correct course/module, and made available for students to stream.

Main Flow of Events:

1. Teacher navigates to a course and selects 'Upload Video'.
2. Teacher provides the title, description, thumbnail, and selects the target module/chapter.
3. System uploads the video file to cloud/file storage and generates metadata (duration, chapter markers).
4. System runs the video through the content moderation queue before publishing.
5. Once approved, the system publishes the video and makes it available for streaming to enrolled students.
6. Teacher can later edit the video's details or replace the uploaded file or delete the video from the platform.

Extensions / Alternative Flows:

- 3a. Upload fails or file format is unsupported: system displays an error and allows the teacher to retry.
- 4a. Moderation flags the video (e.g., banned keyword in the title): video is quarantined, and the teacher is notified instead of being published.



6. Quiz & Assessment Subsystem

Assigned to: Adikari A. M. T. D. (IT25100859)

Use Case Name	Take Quiz
Primary Actor(s)	Student
Secondary Actor(s)	Teacher / Instructor (defines the quiz), System (auto-grading)
Precondition	The student is registered, logged in, enrolled in the course, and the teacher has published a quiz linked to a video/module.
Trigger	The student selects a quiz from an enrolled course.
Postcondition	The student's answers are submitted, a score is calculated and recorded, and the result is available to both student and teacher.

Main Flow of Events:

1. Student navigates to the course and selects an available quiz.
2. System displays the quiz instructions, number of questions, and the timer duration.
3. Student clicks 'Start Quiz'; the system starts the countdown timer and loads the question bank.
4. Student answers each question (multiple choice / short answer) and moves through the quiz.
5. Student submits the quiz before the timer expires.
6. System automatically grades the submission against the answer key and calculates the score.
7. System compares the score against the configured pass mark and records a Pass/Fail status.
8. System displays the score and feedback to the student and stores the result in the database.

